

Commutative Algebra in Mixed Characteristic

1 Lecture 1 — Rowlee Lecture: Bhargav Bhatt

p -adic numbers:

1. $p \rightarrow p$ -adic absolute value (p fixed prime)

$$\left| \frac{a}{b} \right|_p = \left| p^r \cdot \frac{a'}{b'} \right|_p = \frac{1}{p^r} \quad \text{Ex: } p^n \rightarrow 0 \text{ as } n \rightarrow \infty$$

where a', b' are coprime to p .

Definition 1 (Hensel 1897). $\mathbb{Q}_p$: completion of $(\mathbb{Q}, |\cdot|_p)$; can $+$, $-$, $\times$. We have $|\cdot|_p$ on $\mathbb{Q}_p$.

$$\mathbb{Q}_p = \left\{ \sum_{i=-N}^{\infty} a_i p^i \mid a_i \in \{0, \dots, p-1\} \right\} \quad (\text{“Formal Laurent series in } p\text{”})$$

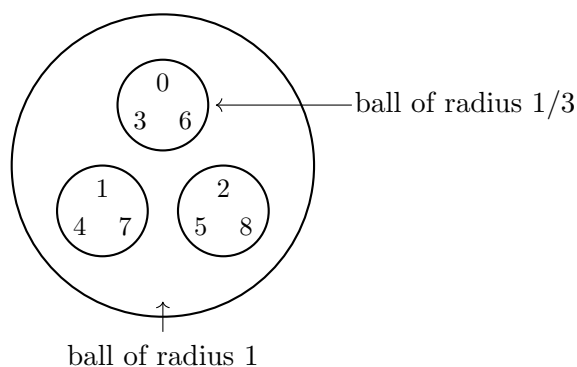
Key Property: For $a, b \in \mathbb{Q}_p$, $|a+b|_p \leq \max(|a|_p, |b|_p)$.

$$\mathbb{Z}_p = \{a \in \mathbb{Q}_p \mid |a|_p \leq 1\} \quad (\text{stable under } +, \times).$$

$\hookrightarrow$ PID, with unique prime (p) .

$$\mathbb{Z}/p\mathbb{Z} \xleftarrow{\text{kill } (p)} \mathbb{Z}_p \xrightarrow{\text{invert } p} \mathbb{Q}_p$$

[Picture of \$\mathbb{Z} \subset \mathbb{Z}_3\$:](#)



Examples:

i) checking integrality:

$$f(x) = \sqrt{1+x} = (1+x)^{1/2} = \sum_{i=0}^{\infty} \binom{1/2}{i} x^i \in \mathbb{Q}[[x]]$$

where,

$$\binom{t}{i} = \frac{t(t-1) \cdots (t-i+1)}{i!}$$

Claim. All coeffs lie in $\mathbb{Z}[1/2]$.

$\Rightarrow$ Fix p odd; so, $1/2 \in \mathbb{Z}_p$. We have a poly map,

$$g : \mathbb{Q}_p \xrightarrow{t \mapsto \binom{t}{i}_{\mathbb{Q}_p}} \mathbb{Q}_p$$
$$g(\mathbb{Z}) \subset \mathbb{Z} \subset \mathbb{Z}_p.$$

By p -adic continuity,

$$g(\mathbb{Z}_p) \subset \mathbb{Z}_p \implies \binom{1/2}{i} \in \mathbb{Z}_p \implies p \nmid \text{denominator of } \binom{1/2}{i}.$$

2) Finding Solutions:

Theorem 1 (Hasse–Minkowski, 1920s). *Let $f(x_1, \dots, x_n)$ be a quadratic form over $\mathbb{Q}$. Then f has a nontrivial zero over $\mathbb{Q}$ if and only if f has a nontrivial zero over $\mathbb{Q}_p$ for every prime p and over $\mathbb{R}$.*

Compare (Selmer 65): Analog of $\Leftarrow$ fails for $3x^3 + 4y^3 + 5z^3$.

3) Counting Solutions:

X = system of poly eqns in n -variables/ $\mathbb{F}_p$.

Theorem 2 (Weil, 1949). *Let X be a variety over $\mathbb{F}_p$. Its zeta function is*

$$Z(X, t) = \exp \left(\sum_{m \geq 1} \#X(\mathbb{F}_{p^m}) \frac{t^m}{m} \right).$$

Then $Z(X, t)$ is a rational function.

- 1) $Z(X, t)$ is a rational function. (Dwork, Grothendieck)
- 2) We know a great deal about the location of its zeros and poles. (Deligne)

Recent Developments: $\mathbb{Q}_p \approx \mathbb{F}_p((t))$

Fontaine-Wintenberger 1979:

$$K = \mathbb{Q}_p(p^{1/p^\infty}) \quad K^\flat = \mathbb{F}_p((t^{1/p^\infty}))$$

$$\mathrm{Gal}(\overline{K}/K) = \mathrm{Gal}(\overline{K^\flat}/K^\flat)$$

Scholze 2012: $\exists$ categories of “perfectoid spaces” over K & $K^\flat$ which are naturally identified.

$$\begin{array}{ccc} \{\text{alg varieties}/K\} & & \{\text{alg varieties}/K^\flat\} \\ \downarrow \text{ART.FORM.} & & \downarrow \text{extract power roots} \\ \mathrm{Perf}_K & \xrightarrow{\quad \simeq \quad} & \mathrm{Perf}_{K^\flat} \end{array}$$

Theorem 3 (André 2016). *Let $R = \mathbb{Z}[x_1, \dots, x_n]$. Suppose $R \xrightarrow{d} S$ is an injective map with S a module finite $/ R$. $\Rightarrow \exists R$ -linear map $\beta : S \rightarrow R$ s.t $\beta \circ d = \mathrm{id}$.*

Theorem 4 (C-A-K-N- 2024).

Theorem 5 (Antieau–Krause–Nikolaus, 2024). *For fixed n , the groups $K_{2i}(\mathbb{Z}/p^n)$ vanish for all sufficiently large i .*

2 Lecture 2 — Connectedness of the associated graded ring- De Stefani

Let $(R, \mathfrak{m}, k)$ be a complete local ring. We can write $R = T/I$ where $T = V[[x_1, \dots, x_n]]$ and V is an unramified DVR with uniformizer $p = \mathrm{char}(k)$. For example, $V = \mathbb{Z}_p$.

$$\begin{array}{c} & k \\ & \swarrow \\ V & \\ & \searrow \\ & V_{(p)} \end{array}$$

The associated graded ring is defined as:

$$G = \mathrm{gr}_{\mathfrak{m}_R}(R) = \mathrm{gr}_{\mathfrak{m}_T}(T)/\mathrm{in}(I) = \bigoplus_{i \geq 0} \mathfrak{m}_R^i / \mathfrak{m}_R^{i+1}$$

where $\mathrm{gr}_{\mathfrak{m}_T}(T) \cong k[X, \dots] = P$.

For $0 \neq f \in T$, if $f \in \mathfrak{m}^v \setminus \mathfrak{m}^{v+1}$, let $v = v(f)$ be the $\mathfrak{m}$ -adic order of f . We define $\mathrm{in}(f)$ as the image of f inside $\mathfrak{m}_T^v / \mathfrak{m}_T^{v+1}$. Then $\mathrm{in}(I) = (\mathrm{in}(f) \mid 0 \neq f \in I)$.

Example 1. $R = k[[t^6, t^7, t^{15}]] \cong \frac{k[[X, Y, Z]]}{(XZ - Y^3, Z^2 - X^5)}$

$$G = \frac{k[X, Y, Z]}{(XZ, Z^2, Y^3Z, Y^6)}$$

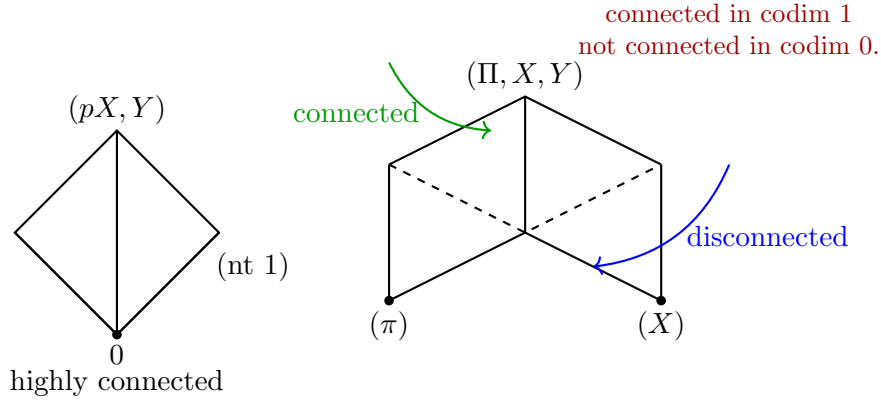
Example 2. $R = \frac{\mathbb{Z}_p[[X, Y]]}{(pX - Y^3)} \quad G = \frac{\mathbb{F}_p[[\Pi, X, Y]]}{(\Pi X)}$

Properties of localized graded rings

Property of localized graded rings	$G \rightsquigarrow R$	$R \rightsquigarrow G$
Regular	✓	✓
Complete Intersection	✓	× (1)
Gorenstein	✓	× (1)
Cohen-Macaulay	✓	× (1)
Reduced	✓	× (1)
Domain	✓	× (2)

Example (2) revisited: For $R = \frac{\mathbb{Z}_p[[X,Y]]}{(pX-Y^3)}$, we localize G :

$$G_{\mathfrak{m}} \cong \left(\frac{\mathbb{F}_p[\Pi, X, Y]}{(\Pi X)} \right)_{(\Pi, X, Y)}$$



Definition 2 (Hartshorne '62). Let A be a ring with $d = \dim(A) < \infty$. A is connected in codim s ($0 \leq s \leq d$) $\iff \text{Spec}(A) \setminus Z$ is connected $\forall Z \subseteq \text{Spec}(A)$ closed with $\dim(Z) < d - s$.
 $[\emptyset$ -disconnected, $\dim(\emptyset) = -1]$

Easy remarks:

- A connected in codim $d \iff \text{Spec}(A)$ connected.
- $(A, \mathfrak{m})$ local, connected in codim $d - 1 \iff \text{Spec}(A) \setminus \{\mathfrak{m}\}$ connected.
- A connected in codim 0 $\iff A_{\text{red}} = A/\sqrt{0}$ domain (remove minimal primes).

Theorem 6 (Hartshorne '62). $(A, \mathfrak{m})$ satisfying Serre's $(S_2) \implies A$ connected in codim 1.

Remark 1. A is connected in codim 1 $\implies A$ equidimensional.

Theorem 7 (Hochster-Huneke '02). $(A, \mathfrak{m})$ complete equidimensional. A connected in codim 1 $\iff H_{\mathfrak{m}}^d(A)$ indecomposable.

”The affine case”

Let $P = k[x_1, \dots, x_n]$, $J \subseteq P$, and let $<$ be a monomial order. Let $\text{in}_<(J)$ be the monomial ideal in P .

Conjecture 8 (Kredel-Weispfenning '88). $P/J \text{ domain} \implies P/\text{in}_<(J) \text{ equidimensional}$.

Theorem 9 (Kalkbrener-Sturmfels '95). Let $k = \bar{k}$. $P/J \text{ domain} \iff P/\text{in}_<(J) \text{ connected in codim 1} \implies P/J \text{ connected in codim 0}$.

Theorem 10 (Varbaro '09). Let $s > 0$. $P/J \text{ connected in codim } s \implies P/\text{in}_<(J) \text{ is connected in codim } s$.

Theorem 11 (De Stefani, Ross, Varbaro). Let $(R, \mathfrak{m})$ be complete, $s > 0$. $R \text{ is connected in codim } s \implies \text{so is } G = \text{gr}_{\mathfrak{m}}(R)$.

Ingredients for Varbaro's thm ['09]:

$$\begin{array}{ccc} k[t] & \longrightarrow & \text{hom}(P/J) \xrightarrow{t \neq 0} P/J \\ & & \searrow t=0 \\ & & P/\text{in}_<(J) \end{array}$$

Theorem 12 (Grothendieck). Let $(A, \mathfrak{m})$ be complete, $s > 0$, and let $t \in \mathfrak{m}$ be a parameter. $A \text{ connected in codim } s \implies \text{so is } A/(t)$.

Homogenization and Orders

Let $T = V[[x_1, \dots, x_n]] \ni f \neq 0$. We have $\text{in}(f) = (x)$ -adic order of f . Over (V, p) , the maximal ideal is $\mathfrak{m}(T) = (\pi, x)$ -order of f .

We construct:

$$T = \frac{S[[\pi]]}{(\pi - p)} \quad S[[t]] = T[[\pi, t]] = B$$

For $F \in S$, $\text{hom}(F) = t^{-\text{in}(F)} \cdot F(x_1 t, \dots, x_n t)$.

Example 3. Let $T = \mathbb{Z}_p[[X, Y]]$ and $I = (pX - Y^3) \ni f$. Let $S = \mathbb{Z}_p[[\pi, X, Y]]$ and $B = S[[t]]$. Then:

$$\text{hom}(f) = t^{-3}(pXt - (Yt)^3) = pX - Y^3 t^2$$

Starting with R :

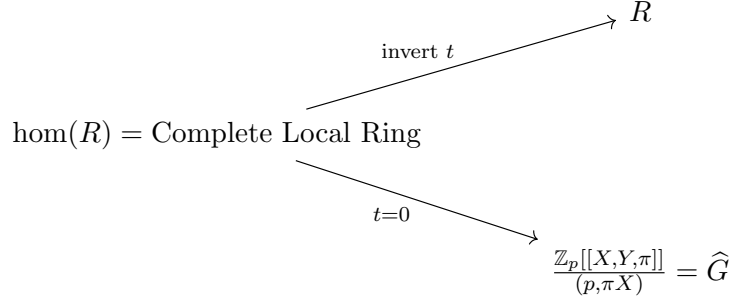
$$R = \frac{\mathbb{Z}_p[[X, Y]]}{(pX - Y^3)} \rightsquigarrow R = \frac{\mathbb{Z}_p[[\pi, X, Y]]}{(\pi - p, pX - Y^3)}$$

Let $I' = (\pi - p, pX - Y^3)$. We define $\text{hom}(I') = (\text{hom}(f) \mid 0 \neq f \in I')$.

$$\begin{aligned} \text{hom}(\pi - p) &= \pi t - p \\ \text{hom}(pX - Y^3) &= pX - Y^3 t^2 \\ \text{hom}(\pi X - Y^3) &= \pi X - Y^3 t \end{aligned}$$

So the homogenization of R is:

$$\text{hom}(R) = \frac{\mathbb{Z}_p[[X, Y, \pi, t]]}{\text{hom}(I')} = \frac{\mathbb{Z}_p[[X, Y, \pi, t]]}{(\pi t - p, pX - Y^3 t^2, \pi X - Y^3 t)}$$



Theorem 13. *Let $(R, \mathfrak{m})$ be complete over V . Then:*

$$\text{hom}(R) \widehat{\otimes}_{V[[t]]} V((t)) \cong R \widehat{\otimes} V((t))$$

Moreover, if R is connected in codim s , so is $\text{hom}(R)$. The connectedness graph $\Gamma_s(R)$ is connected $\iff \Gamma_s(\text{hom}(R))$ is connected.

3 Lecture 3 — Perfectoid Pure Singularities by Linquan Ma

Let p be a fixed prime number.

Definition 3. Let R be a Noetherian $\mathbb{F}_p$ -algebra. The ring R is defined to be:

- **F-pure** if the Frobenius map $R \xrightarrow{F} R$ is pure as a map of R -modules. (Recall that $R \rightarrow F_* R$ is pure $\iff R \rightarrow F_* R$ splits, when R is F-finite).
- **F-injective** if the induced map on local cohomology $H_{\mathfrak{m}}^i(R) \xrightarrow{F} H_{\mathfrak{m}}^i(R)$ is injective for all $\mathfrak{m} \in \text{MaxSpec}(R)$ and for all i .

Remark 2. • R is F-pure / F-injective $\iff$ so is $R_{\mathfrak{m}}$ for all $\mathfrak{m} \in \text{MaxSpec}(R)$ $\iff$ so is R_Q for all $Q \in \text{Spec}(R)$.

- We have the following implications:

$$\text{F-pure} \implies \text{F-injective} \implies \text{reduced} + \text{semi-normal}$$

Furthermore, if R is Gorenstein, F-injective implies F-pure.

- (Hara-Watanabe) F-pure and $\mathbb{Q}$ -Gorenstein $\implies$ log canonical.

Example 4. (a) Let $R = K[x, y, z]/(f)$ such that $X = \text{Proj}(R)$ is an elliptic curve. Then, R is F-pure $\iff X = \text{Proj}(R)$ is ordinary $\iff H^1(X, \mathcal{O}_X) \xrightarrow{F} H^1(X, \mathcal{O}_X)$ is injective.

(b) Let $R = K[x_1, \dots, x_n]/(\text{sq-free monomial ideal})$. Then R is F-pure. (Fedder's criterion: If $R = S/I$, where S is a regular local graded ring, then R is F-pure $\iff I^{[p]} : I \not\subset \mathfrak{m}^{[p]}$).

(c) Let $R = K[x_1, \dots, x_n]/I$. If $\text{in}_{<}(I)$ is square-free, then R is F-injective but not necessarily F-pure.

Theorem 14 (Fedder, Horiuchi-Miller-Shimomoto). *Let $(R, \mathfrak{m})$ be a Noetherian local $\mathbb{F}_p$ -algebra, and let $f \in \mathfrak{m}$ be a non-zero divisor. Suppose one of the following holds:*

1. $R/(f)$ is Cohen-Macaulay and F -injective.
2. $R/(f)$ is F -pure.

Then, R is F -injective. (In particular, if R is Gorenstein and $R/(f)$ is F -pure, then so is R).

$$\begin{array}{ccc}
& \tilde{R}/t & \xrightarrow{\simeq} K[x]/\text{in}_{>}(I) \\
& \uparrow & \\
K[t] & \hookrightarrow & \tilde{R} \\
& \downarrow & \\
& \tilde{R} \otimes_{K[t]} K(t) & \xrightarrow{\simeq} R \otimes_K K(t)
\end{array}$$

We note the equivalences:

$$R \text{ is } F\text{-pure} \iff R \rightarrow F_* R \text{ is pure} \iff R \rightarrow F_*^e R \text{ is pure } \forall e \iff R \rightarrow \varinjlim_e F_*^e R \text{ is pure.}$$

Here, $\varinjlim_e F_*^e R = R_{\text{perf}}$, which is a perfect ring.

Definition 4 (Bhatt-Morrow-Scholze). A ring S is called **perfectoid** if:

1. S is p -complete and $\exists \varpi \in S$ such that $\varpi^p = p \cdot u$ for some $u \in S^\times$.
2. $S/p \xrightarrow{\text{Frob}} S/p$ is surjective.
3. $W(S^\flat) \xrightarrow{\theta} S$, with $\ker(\theta)$ being a principal ideal.

Here, $S^\flat = \varprojlim_{\text{Frob}} S/p$.

Example 5. 1. Perfect rings of characteristic p . $W(S) \xrightarrow{\text{Pro}} S$, $\ker = (p)$.

2. $\mathbb{Z}_p[x_1, \dots, x_d] \rightarrow \mathbb{Z}_p \left[p^{\frac{1}{p^\infty}}, x_1^{\frac{1}{p^\infty}}, \dots, x_d^{\frac{1}{p^\infty}} \right]^{\wedge p}$ is a perfectoid ring.

(If S is p -torsion free, then S is perfectoid $\iff S/\varpi \xrightarrow{\text{Frob}} S/p$ is bijective).

(3) Let R be a Noetherian complete local domain with $\text{char}(R/\mathfrak{m}) = p > 0$. Then $(R^+)^{\wedge p} \leftarrow$ perfectoid.

Definition 5. Let R be a Noetherian ring such that $p \in J(R)$ (the Jacobson radical of R). Then, R is **perfectoid pure** if $\exists$ a perfectoid R -algebra B so that $R \rightarrow B$ is pure.

(We say R is **perfectoid injective** if $\exists B$ such that $H_{\mathfrak{m}}^i(R) \rightarrow H_{\mathfrak{m}}^i(B)$ is injective for all $\mathfrak{m}$, for all i).

Example 6. 1. In characteristic p , perfectoid pure $\iff$ F -pure, and perfectoid injective $\iff$ F -injective.

2. Regular local rings of mixed characteristic p are perfectoid pure.

$$A = \mathbb{Z}_p[[x_1, \dots, x_d]] \rightarrow A_\infty = A \left[p^{\frac{1}{p^\infty}}, x_1^{\frac{1}{p^\infty}}, \dots, x_d^{\frac{1}{p^\infty}} \right]^{\wedge p} \text{ is faithfully flat.}$$

3. All splinters of mixed characteristic are perfectoid pure. $R \rightarrow R^+$ is pure ($\implies R \rightarrow (R^+)^{\wedge p}$ is pure). E.g., summand of regular rings.

Remark 3. • R is perfectoid pure/injective $\iff$ so is $R_{\mathfrak{m}}$ for all $\mathfrak{m} \in \text{MaxSpec}(R)$ $\iff$ so is R_Q for all $Q \in V(p)$.

- We have the implications:

$$\text{perfectoid pure} \implies \text{perfectoid injective} \implies \text{reduced} + \text{semi-normal}.$$

Also, perfectoid injective + Gorenstein implies perfectoid pure.

Theorem 15 (BMPSTWW). *If R is perfectoid pure and $\mathbb{Q}$ -Gorenstein with index not divisible by p , then R is log-canonical.*

Theorem 16 (BMPSTWW). *Let $(R, \mathfrak{m})$ be a Noetherian local ring, and let $f \in \mathfrak{m}$ be a non-zero divisor. If $R/(f)$ is a complete intersection and perfectoid pure, then R is perfectoid pure.*

Example 7. 1. Let $R = \mathbb{Z}_p[x, y, z]/(x^3 + y^3 + z^3)$. If $p \equiv 1 \pmod{3}$, then R is perfectoid pure. (Proof: $R/p \cong \mathbb{F}_p[x, y, z]/(x^3 + y^3 + z^3)$ is F-pure $\square$).

1'. $R = \mathbb{Z}_p[x, y, z]/(x^3 + y^3 + z^3)$, $p \equiv 2 \pmod{3}$. Yes, R is still perfectoid pure.

Theorem 17 (Yoshikawa). *Let $(R, \mathfrak{m})$ be a Noetherian local p -torsion free ring, which is a complete intersection. If R/pR is quasi-F-split, then R is perfectoid pure. (e.g., Cone of elliptic curve are always quasi-F-split).*

Example 8. 2. Let $R = \mathbb{Z}_p[x, y]/(p^3 + x^2 + y^3)$, with $p \equiv 1 \pmod{3}$. R is perfectoid pure.

$$\underbrace{\text{gr}_{\mathfrak{m}}(R)}_{\text{F-pure}} \rightsquigarrow \underbrace{R[\text{mt}, t^{-1}]_{(\mathfrak{m})}}_{\text{perfectoid pure}} \rightsquigarrow R \text{ is a summand of } R[\text{mt}, t^{-1}]_{(\mathfrak{m})} \implies R \text{ is perfectoid pure. } \square$$

2'. Let $R = \mathbb{Z}_p[x, y]/(p^3 + x^3 + y^3)$, with $p \equiv 2 \pmod{3}$. R is not perfectoid pure. (Brozzo, Tagathost-Pandey-Moreno-Schwede-Sridhar).

Question: Is there a Fedder type criterion for perfectoid pure singularities?

4 Lecture 4: Are Determinantal Rings Pure Subrings of Polynomial Rings? By Anurag Singh

Definition 6. A ring homomorphism $\phi: R \rightarrow S$ is **PURE** if $(R \rightarrow S) \otimes_R M$ is injective $\forall$ R -modules M .

- $R \hookrightarrow S$ faithfully flat $\implies$ pure.
- Let V be a finite rank vector space over $\mathbb{C}$.
- Let G be a finite subgroup of $\text{GL}(V) \curvearrowright S = \text{Sym}(V)$.

We have that $S^G \hookrightarrow S$ is faithfully flat $\iff S^G$ is a polynomial ring. However, the inclusion $S^G \hookrightarrow S$ is always pure.

- If $R \hookrightarrow S$ is R -split, then it is pure.
- If R is a complete local ring, or if R, S are $\mathbb{N}$ -graded finitely generated with $R_0 = S_0$ a complete local ring, then $R \hookrightarrow S$ is R -split $\iff$ pure.

Suppose R is an $\mathbb{N}$ -graded finitely generated ring over a field $R_0 = K$, and R is a pure subring of a polynomial ring $K[X]$. Then:

- (a) R is normal.
- (b) (Hochster-Roberts) R is Cohen-Macaulay.
- (c) (Boutot) In characteristic 0: R has rational singularities.
- (d) (Hochster-Huneke) R is F -regular.
- (e) (Smith) R is simple as a $D_{R|K}$ -module.
- (f) (Smith-Van den Bergh) In characteristic p :

$$\begin{cases} R \text{ has FFRT} \\ D_{R|K} \text{ is a simple ring} \end{cases}$$

- (g) (Huneke-Sharp, Lyubeznik) Fix an ideal I in R . Then $\text{Ass}_R H_I^k(R)$ is finite.

Theorem 18 (Heitmann-Ma). *For mixed characteristic $R \hookrightarrow S$, if it is pure with S regular, then R is pseudo-rational (in particular, Cohen-Macaulay).*

If G is a linearly reductive subgroup of $\text{GL}_n(K)$, acting on a polynomial ring $S = K[X]$, then $S^G \hookrightarrow S$ is pure.

Determinantal Rings

Let $G = \text{GL}_t(\mathbb{C}) \curvearrowright S = \mathbb{C}[Y_{m \times t}, Z_{t \times n}]$. The action is given by:

$$\begin{aligned} M: Y &\mapsto YM^{-1} \\ Z &\mapsto MZ \end{aligned} \implies YZ \mapsto YM^{-1}MZ = YZ \quad [m, n \geq t+1, t \neq 0]$$

This gives the isomorphism and inclusions:

$$\frac{\mathbb{C}[X_{m \times n}]}{I_{t+1}} \simeq \mathbb{C}[YZ] = \mathbb{C}[Y, Z]^{\text{GL}_t} \subseteq \mathbb{C}[Y, Z]$$

Symmetric Determinantal Rings

Let $G = O_t(\mathbb{C}) = \{M \in \text{GL}_t : M^t M = \text{id}\}$. The action is given by $M: Y \mapsto MY$, so $Y^t Y \mapsto Y^t M^t M Y = Y^t Y$.

$$\begin{aligned} \mathbb{C}[Y]^{O_t} &\hookrightarrow \mathbb{C}[Y_{t \times n}] \\ \parallel & \\ \mathbb{C}[Y^t Y] & \qquad \qquad [n \geq t+1, t \neq 0] \\ \simeq \downarrow & \\ \frac{\mathbb{C}[X_{n \times n}^{\text{Sym}}]}{I_{t+1}} & \end{aligned}$$

Pfaffian Determinantal Rings

Let $G = \mathrm{Sp}_{2t}(\mathbb{C}) = \{M \in \mathrm{GL}_{2t} : M^t \Omega M = \Omega\}$, where $\Omega = \begin{pmatrix} 0 & -\mathrm{id}_t \\ \mathrm{id}_t & 0 \end{pmatrix}$. The action is given by $M : Y \mapsto MY$. Thus, $Y^t \Omega Y \mapsto Y^t M^t \Omega M Y = Y^t \Omega Y$.

$$\frac{\mathbb{C}[X_{n \times n}^{\mathrm{alt}}]}{\mathrm{Pf}_{2t+2}} \simeq \mathbb{C}[Y^t \Omega Y] = \mathbb{C}[Y]^{\mathrm{Sp}_{2t}} \subseteq \mathbb{C}[Y] \quad [n \geq 2t+2, t \neq 0]$$

Theorem 19 (Hochster-Jeffries-Pandey-Singh). *Let K be a field of characteristic $p > 0$.*

1. $K[YZ] \subseteq K[Y, Z]$ is pure $\iff t = 1$.
2. $K[Y^t Y] \subseteq K[Y]$ is pure $\iff \begin{cases} t = 1 & \text{or} \\ t = 2, & p \text{ odd} \end{cases}$
3. $K[Y^t \Omega Y] \subseteq K[Y]$ is pure $\iff$ Pigs Fly (?)

Question: Is the LHS a pure subring of SOME polynomial ring?

Theorem 20 (Jeffries, Singh). *Assume $m, n \geq t+1$, and let V denote $\mathbb{Z}$ or $\widehat{\mathbb{Z}_{(p)}}$. Then:*

1. $V[YZ]$ is a pure subring of some polynomial ring over V if and only if $t = 1$.
2. $V[Y^t Y]$ is a pure subring of some polynomial ring over V if and only if $t = 1$, or $t = 2$ and $V = \widehat{\mathbb{Z}_{(p)}}$ for p odd.
3. $V[Y^t \Omega Y]$ is a pure subring of some polynomial ring over $V \iff$ Pigs Fly.

Let Y be a $2 \times n$ matrix.

$$\frac{\mathbb{Z}[Y^t Y]}{I_3} \simeq \frac{\mathbb{Z}[X_{n \times n}^{\mathrm{Sym}}]}{I_3} \simeq \mathbb{Z}[Y^t P Y] \quad \text{where } P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Notice that $(Y^t P Y)^t = Y^t P^t Y = Y^t P Y$. Let $Y = \begin{pmatrix} y_1 & \cdots & y_n \\ z_1 & \cdots & z_n \end{pmatrix}$. Then:

$$Y^t P Y = \begin{pmatrix} y_1 & z_1 \\ \vdots & \vdots \\ y_n & z_n \end{pmatrix} \begin{pmatrix} z_1 & \cdots & z_n \\ y_1 & \cdots & y_n \end{pmatrix} = \begin{pmatrix} 2y_1 z_1 & y_1 z_2 + y_2 z_1 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

Now consider the rings over $\mathbb{Z}[\frac{1}{2}]$:

$$\mathbb{Z}\left[\frac{1}{2}\right][Y^t P Y] = \mathbb{Z}\left[\frac{1}{2}\right][y_i z_i, y_i z_j + y_j z_i] \subseteq \mathbb{Z}\left[\frac{1}{2}\right][y_i z_j] \subseteq \mathbb{Z}\left[\frac{1}{2}\right][y_i, z_i]$$

Here, the group $S_0 = \langle \sigma \rangle \cong \mathbb{Z}/2\mathbb{Z}$ acts via $\sigma : y_i \leftrightarrow z_i$. The subring $\mathbb{Z}[\frac{1}{2}][y_i z_j]$ corresponds to S_0 -invariants, and $\mathbb{Z}[\frac{1}{2}][y_i, z_i]$ is the polynomial ring S . The inclusion $\mathbb{Z}[\frac{1}{2}][y_i z_j] \subseteq \mathbb{Z}[\frac{1}{2}][y_i, z_i]$ is pure (for degree reasons).

Conclusion: All inclusions are pure.